



Engineering the facilities advanced industry runs on.

EPCM, EPCC and Energy Facility Management for cleanroom-critical industries: semiconductor, data centre, EV battery, pharmaceutical and energy. Established 1994.

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PART 01 OF 03

01

The Company

SECTIONS 01.1 TO 01.9

Established 1994 · 32 years in total facility solutions

450 people across 6 global offices

ISO 9001 · ISO 14001 · ISO 45001 certified



About IAQ

Who we are and what we build. The delivery models are in 01.7.

Established 1994. A trailblazer in total facility solutions.

Engineering, procurement, construction and maintenance for the industries where contamination, uptime and safety decide the outcome, from semiconductor wafer fabs to gigafactories, data centres and pharmaceutical plants.

Vision

To be a regional facility solutions provider with engineering excellence that facilitates technological innovation and advancement in quality of life.

Mission

Providing innovative, sustainable facility and engineering solutions benefitting our clients and stakeholders, driven by our leadership, employees and partners globally.

IAQ at a Glance

The company in six numbers, and the certifications that stand behind them.



31

YEARS EXPERIENCE



450

EMPLOYEES



200+

PROJECTS COMPLETED



1.5 mil m²

CLEANROOM BUILT-UP AREA



7

GLOBAL OFFICES



20 t / yr

CARBON FOOTPRINT REDUCED

CERTIFICATION & RECOGNITION

INTERTEK **ISO 9001:2015**

INTERTEK **ISO 14001:2015**

INTERTEK **ISO 45001:2018**

CIDB **GRADE G7**

PKK **GRADE G7**

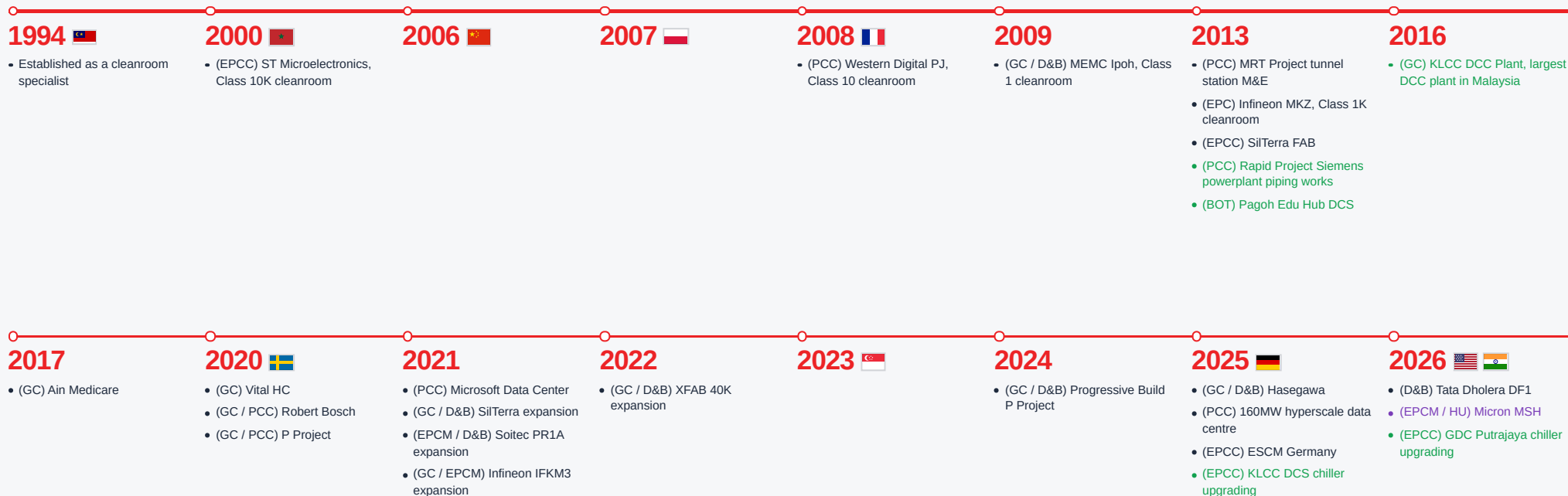
HIGHWIRE SAFETY **GOLD AWARD**

MCIEA 2024 **BUILDER OF THE YEAR**

Milestones

IAQ's own timeline, 1994 to 2026: the projects that marked each year and the offices opened along the way.

● EPCC PROJECTS ● ENERGY PROJECTS ● TOOL UTILITY PROJECTS ● OFFICE OPENED



Global Footprint

Seven markets, from the Malaysian headquarters outward.

**1994****Malaysia**

Turnkey design, procurement, construction and commissioning for hi-tech clients. Selangor (HQ), Penang, Johor and Kuching.

**2020****Sweden**

Gigafactory construction for EV battery manufacture, part of Europe's clean and digital transition.

**2023****Singapore**

Design & build EPCM services to global clients in the semiconductor industry.

**2025****Germany**

EPCC services following the semiconductor industry expansion in Europe.

**2026****India**

D&B turnkey delivery for India's first wafer fab.

**2026****United States**

Entry to the US following the localisation of advanced technology facilities.

**—****Ireland**

Newest market. Scope and year to be confirmed by IAQ.

Core Values

Six commitments that govern how every project is run. Not words, but working rules.



01

Safety First

Our services and works are carried out to the highest standard of safety and ethics.



02

Quality Consistency

We take pride in the quality of solutions delivered, to achieve maximum client satisfaction.



03

Honesty and Integrity

Our core code of conduct, fostering trust, accountability, professionalism and ethical practice.



04

Efficiency & Proficiency

Committed to maximising available resources to achieve the best result collectively.



05

Engineering Capabilities

Our way of working revolves around engineering principles, developing precise and sustainable solutions.



06

Pursuit of Excellence

Devoted to excellence in all our works, providing sustainable solutions to complex challenges.

Business Units

Three delivery capabilities under one group, covering the full chain.



01

Engineering, Procurement and Construction (EPC)

EPC from conception to operation, completion and maintenance. Every stage managed, from initial design through commissioning, so clients in hi-tech industries can bring their visions to life.



02

Process Critical Utilities & Total Tool Install Solutions

EPCM partner for semiconductor manufacturing: process-critical utility infrastructure and total tool installation, bridging facility readiness and manufacturing start-up to accelerate fab ramp-up.



03

Energy Management

Energy management solutions that optimise operations and reduce carbon footprint, keeping facilities running at the highest levels of efficiency and sustainability.

Delivery Models

The three models IAQ contracts under, across the full life of a facility.

EPCM

Engineering, Procurement & Construction Management

Management of engineering, procurement and construction for semiconductor manufacturing, bridging facility readiness and manufacturing start-up to accelerate fab ramp-up.

DELIVERED UNDER THIS MODEL

- SilTerra expansion
- Soitec PR1A expansion
- Infineon IFKM3 expansion
- Micron MSH

EPCC

Engineering, Procurement, Construction & Commissioning

Full turnkey delivery from conception to operation, completion and maintenance, with every stage managed from initial design through commissioning.

DELIVERED UNDER THIS MODEL

- ST Microelectronics
- SilTerra FAB
- ESCM Germany
- KLCC DCS chiller upgrading

EFM

Energy Facility Management

Ongoing energy and facility management that optimises operations and reduces carbon footprint, keeping facilities at the highest levels of efficiency.

DELIVERED UNDER THIS MODEL

- KLCC DCC Plant
- GDC Putrajaya chiller upgrading
- Pagoh Edu Hub DCS

Scope of Services

Twelve packages, self-performed and integrated under one contract.



01

Civil, Structural & Architectural

Greenfield and brownfield CSA



02

Air Conditioning System

ACMV, make-up air, recirculation



03

HT & LV Electrical System

Up to 33kV substations



04

Fire Protection System

Detection, suppression, compliance



05

Cleanroom System

ISO 3 to ISO 8 · Class 1 to Class 100K



06

Process Utilities

CDA, PCW



07

Process Exhaust System

Scrubbed, heat and general exhaust



08

Chemical / Gas Delivery System

Specialty and bulk gases, chemical delivery



09

Ultra Pure Water (UPW)

UPW generation and distribution



10

Waste Water Treatment

Industrial effluent and recovery



11

Facility Monitoring & Control System

FMCS, BMS, PA



12

Tools Hookup

Progressive tool install and hookup

Industry Focus

Seven sectors, one discipline: contamination control under code and class.



01

Semiconductor

Wafer fab, advanced packaging, test & assembly



02

Data Centre

Hyperscale cooling, security, uptime



03

EV Battery

Dry rooms, moisture control, explosion proofing



04

District Cooling & Heating

Urban energy plant and distribution



05

Photovoltaics

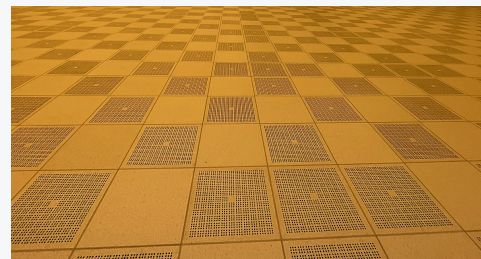
Cell and module lines, toxic material handling



06

Pharmaceuticals & Hospitals

GMP grades, regulated environments



07

Food & Beverage

Hygienic design, quality and safety standards



+

Your facility next

Total facility solutions, end to end.

PART 02 OF 03

02

Project References

SECTIONS 02.1 TO 02.11

200+ projects completed across three regions

Over 1.5 million m² of cleanroom built

ISO 3 to ISO 8 · Class 1 to Class 100K



Portfolio Overview

How the reference book breaks down by region, and which sections cover what.



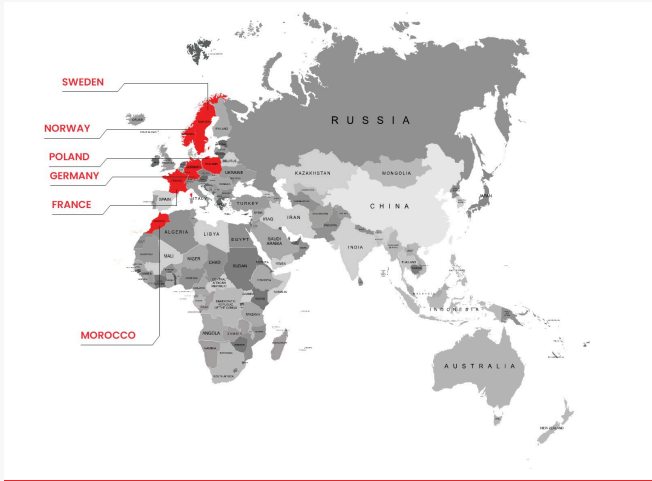
02.2 TO 02.4 · 02.8 TO 02.10

Malaysia & Singapore

Semiconductor, data centre, pharmaceutical, photovoltaics, district cooling and energy.

200+

PROJECTS COMPLETED



02.5 TO 02.6

Europe & Morocco

EV battery gigafactories, wafer fabs and advanced semiconductor in Sweden, Norway, France, Poland and Morocco.

1.5 mil m²

CLEANROOM BUILT



02.7

China

Semiconductor, display, pharmaceutical, automotive and R&D facilities.

5

CONTRACT MODELS

GC · EPCC · EPCM · PCC · Tool Hookup

Semiconductor · Malaysia

Wafer fab, advanced packaging, test and assembly. Further Malaysian projects are listed in 02.11.



KULIM, KEDAH · GREENFIELD WAFER FAB

Infineon Kulim Wafer Fab 3

DESCRIPTION	General contractor for WP06, KLM3 Expansion, 35,000 m² cleanroom area
SCOPE	PCC for cleanroom and mechanical works
CLEANROOM	ISO 4 · 5 · 6 · 7 (CLASS 10 · 100 · 1K · 10K)



ADVANCED PACKAGING · 75,000 M²

P Project

DESCRIPTION	Progressive-built EPCM contract model for the client's advanced packaging facility
SCOPE	Design & build GC: CSA, MEP, process utilities, chemical & gases system, waste treatment
CLEANROOM	ISO 4 TO 7 (CLASS 10 TO 10K)

Semiconductor · Singapore

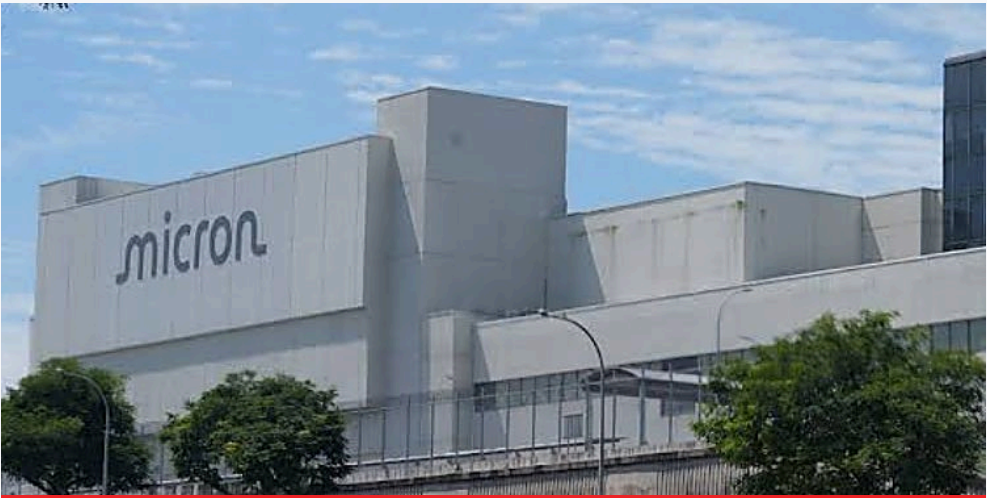
Design & build EPCM in Singapore since 2023.



PHOTOMASK FACILITY

Teksend Photomask

DESCRIPTION	Addition and alteration to an existing two-storey building with cleanroom facilities and ancillary office
SCOPE	EPCC for cleanroom, ACMV, electrical, process utility, BMS, PA, WWT, UPW, tool hookup
CLEANROOM	CLASS 1 TO CLASS 10K



PROGRESSIVE TOOL HOOKUP

Micron MSH Tool HU

DESCRIPTION	2,000 progressive tool hookup services for MSH
SCOPE	Progressive tool hook up
CLEANROOM	Not applicable

Data Centre

Hyperscale and colocation, 9.6 MW to 160 MW.



CLIENT CONFIDENTIAL · 160 MW

Hyperscale Data Centre

DESCRIPTION	Office, data halls and mechanical utility building
SCOPE	Construction, completion, testing & commissioning for mechanical packages 1 & 2
CLEANROOM	Not applicable



KUL03 PHASE 1 · 9.6 MW

Microsoft DTC-KUL 03

DESCRIPTION	9.6 MW data centre
SCOPE	PCC for CHW and CW piping works
CLEANROOM	Not applicable

EV Battery · Europe

Dry rooms for the gigafactories: moisture control and contamination control at scale.



SKELLEFTEÅ, SWEDEN · 62,000 M²

Northvolt AB Gigafactory

DESCRIPTION	Europe gigafactory for lithium-ion battery manufacture
SCOPE	Cleanroom and dry room architecture works
CLEANROOM	Not applicable



NORWAY · BATTERY COAST · 8,600 M²

Morrow Batteries

DESCRIPTION	Norway's Battery Coast project
SCOPE	Cleanroom and dry room architectural system, ACMV
CLEANROOM	Not applicable

Semiconductor · Europe & Morocco

Following the fabs into Europe: advanced-node cleanrooms and progressive hookup.



CLIENT CONFIDENTIAL · WAFER FAB

28/22nm CMOS · 16/12nm FinFET Plant

DESCRIPTION	Design, supply, installation & commissioning of cleanroom facilities
SCOPE	Cleanroom & mechanical general contractor
CLEANROOM	CLASS 1 TO CLASS 10K



CASABLANCA, MOROCCO · 60,000 M²

STMicroelectronics

DESCRIPTION	Built-up area up to 60,000 m²
SCOPE	Cleanroom, mechanical, electrical, process utilities and hookup works
CLEANROOM	ISO 5 (CLASS 100)

Semiconductor & Display · China

EPCC across semiconductor, display and precision manufacturing.



HANGZHOU · 23,400 M²

Ferrotec Semiconductor

DESCRIPTION	Built-up area up to 23,400 m ²
SCOPE	EPCC for cleanroom, mechanical, electrical and plumbing works and hookup
CLEANROOM	ISO 4 · 5 (CLASS 10 & 100)



WUXI · 8,000 M²

Infineon Technologies

DESCRIPTION	Built-up area up to 8,000 m ²
SCOPE	EPCC for cleanroom, mechanical, electrical and plumbing works and hookup
CLEANROOM	ISO 5 (CLASS 100)

Photovoltaics

Solar cell and module manufacturing, including toxic material and waste handling.



KULIM, MALAYSIA

First Solar Malaysia

DESCRIPTION	First Solar KMW building in Kulim
SCOPE	PCC for mechanical & electrical, KMW building; CSA and M&E for forming gas plant; M&E for SCO1
CLEANROOM	Not applicable



CHINA · 57,000 M²

Chint Solar Technology

DESCRIPTION	Built-up area up to 57,000 m ²
SCOPE	Cleanroom system, ACMV, process utilities and tools hookup
CLEANROOM	ISO 8 (CLASS 100K)

Pharmaceutical & Medical

GMP grades and regulated environments, from parenteral facilities to medical device plants.



GREENFIELD · MEDICAL DEVICE

Insulet

DESCRIPTION	Electronic medical device manufacturing plant, greenfield project
SCOPE	Design and build mechanical, cleanroom & electrical package
CLEANROOM	ISO 8 (CLASS 100K)



PUCHONG, SELANGOR

Pharmaniaga

DESCRIPTION	Small volume parenteral facility
SCOPE	Cleanroom system, air conditioning and mechanical ventilation
CLEANROOM	ISO 5 · 7 · 8 (CLASS 100 · 10K · 100K)

District Cooling & Energy

Plant, chillers and co-generation for urban and industrial energy.



GAS DISTRICT COOLING (M) SDN BHD

KLCC District Cooling Plant

DESCRIPTION	LARGEST DISTRICT COOLING CENTRE IN MALAYSIA
SCOPE	ACMV, electrical & instrumentation, fire protection, process utility, plumbing & sanitary, water treatment, CSA
CLEANROOM	Not applicable



PICC PLANT, PUTRAJAYA

GDC Putrajaya

DESCRIPTION	For the Putrajaya International Convention Centre (PICC) plant
SCOPE	EPCC of plant electrification, chiller replacement and associated works at Gas District Cooling Putrajaya
CLEANROOM	Not applicable

Reference Index

A selection from 200+ completed projects. The full reference list is available on request.

MALAYSIA · SEMICONDUCTOR

XFAB Kuching 40K Expansion, 40,000 m²

SOITEC PR1A Expansion, 45,000 m²

Robert Bosch Testing Manufacturing Plant

Wafer Fab Facility & Expansion, FAB1E A/B

Texas Instruments, Melaka, 32,000 m²

Infineon Technologies, Melaka, Block 8

MEMC Ipoh, Class 1 cleanroom

STMicroelectronics, Johor

Western Digital (M), Petaling Jaya

WD Media (M), Penang, MA 28 & MA 29

Flextronics Shah Alam

MALAYSIA · OTHER SECTORS & SINGAPORE

AUO Sunpower, Melaka, Fab 3A

Vital Healthcare, medical device plant

Ain Medicare, pharmaceutical building

M Project, food flavouring plant, 40,000 m²

P Project, MEP works, mega lab retrofit

Rapid Petronas Pengerang Co-gen

Pagoh Education Hub DCS

Rapid MCD chemical plant

Matrix (M) Sdn Bhd

T Hasegawa, spray dryer & microwave tunnel

Elegant Aura (M) Sdn Bhd

Caterpillar Asia, 7T office renovation

Micron F10A & F10NX, Singapore

National University of Singapore, tissue culture lab

EUROPE & MOROCCO

ACC Phase 1, France, 35,000 m²

Envision, France

Soitec Semiconductor, Paris, 11,000 m²

Xfab Semiconductor, Paris

Nemotek, Rabat, Morocco

Kimoto, Poland

Sumika, Poland

CHINA

SMIC, Ningbo, 22,000 m²

Maxtor Technology, Suzhou, 95,000 m²

All-Cent RF Technology, Wuxi, 25,600 m²

Kunshan Visionox Display, 27,000 m²

GS Magicdrive Inc., 68,000 m²

Shantou Goworld Display

M-Flex (Suzhou), 25,000 m²

MMI Industries (Wuxi)

Huawei Technologies Co., Ltd

Shanghai Roche Pharmaceuticals

Suzhou Ascentage Pharma, 25,800 m²

Jiangsu GenScript Biotech

Mindray Group Co., Ltd

Unilever · GE · Dow · Rohm and Haas R&D centres

PART 03 OF 03

03

Safety, Quality & ESG

SECTIONS 03.1 TO 03.4

Highwire Gold safety award

ISO 45001:2018 occupational health & safety

20 tons per annum carbon footprint reduction



ESG Commitments

Three pillars, applied to operations and to every project we take on.



E

Environmental Responsibility

Minimising environmental footprint through sustainable design and construction. Energy-efficient solutions and eco-friendly technologies, prioritised at design stage.



S

Social Accountability

The well-being and safety of employees and communities first. A diverse, inclusive work environment and ethical labour practices.



G

Corporate Governance

Transparency, accountability and ethical decision-making, with the highest standards of compliance, building trust with clients, partners and stakeholders.

Safety & Recognition

How our standards are independently verified and recognised.

Judged on the things that actually matter.

MCIEA award recipients are evaluated across company performance, project management, technical expertise, innovation, quality, safety and sustainability. The same criteria our clients audit us against.

Builder of the Year 2024

Malaysian Construction Industry Excellence Awards (MCIEA), awarded by CIDB.

Highwire Gold

Gold-rated contractor safety performance.

ISO 45001:2018

Occupational health and safety management, certified by Intertek.



Closing

Your Total Facility Solutions Provider.

We infuse safety, quality, efficiency and excellence into every project.

From engineering design and project management to construction and maintenance, across semiconductor manufacturing, clean energy and every environment in between. IAQ is your trusted partner for total facility solutions.

Contacts

For inquiry and business opportunity.



YOUR TOTAL FACILITY SOLUTIONS PROVIDER

Engineering, procurement, construction and energy management for the industries where contamination, uptime and safety decide the outcome. Established 1994.

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